

## INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

### THE SITE

Location NAO 1/NAO 2, OH -- station KCMH, America/New\_York  
Committed pair OSM 671301768 -> 802160675  
NAO 1/NAO 2 / NAO 3  
Facade gap 115.9 m between the two halls  
Weather record 43,786 real hourly records from KCMH  
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.4443 C at 300 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

### THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2024-02-28 (crossing)  
Plant limit 18.0 C  
Notice required 6 h  
Forecast skill 0.00 relative to persistence  
Level anchor sensor  
Condenser bank longest facade  
Switch budget 2 mode changes per day  
Minimum dwell 3 h  
Max dew point 15.0 C (Green Grid WP#46 p.6)

### WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 1 mode change. That is 3 more than the reactive incumbent, at 0 unsafe hours against its 2. 5 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 8 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)  
Incumbent 5 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s)  
5 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

### EVERY HOUR, AND THE REASON FOR IT

| hh | mode       | safe | binding       | bound  | limit | actual | margin |
|----|------------|------|---------------|--------|-------|--------|--------|
| 00 | mechanical | no   | dew point     | 15.809 | 18.0  | 16.110 | 4.078  |
| 01 | mechanical | no   | dew point     | 15.805 | 18.0  | 16.110 | 3.654  |
| 02 | mechanical | no   | dew point     | 16.319 | 18.0  | 16.670 | 3.646  |
| 03 | mechanical | yes  | switch budget | 16.403 | 18.0  | 18.330 | 3.300  |
| 04 | mechanical | yes  | switch budget | 16.319 | 18.0  | 18.890 | 2.873  |
| 05 | mechanical | yes  | switch budget | 16.285 | 18.0  | 17.220 | 2.477  |
| 06 | mechanical | yes  | switch budget | 16.951 | 18.0  | 15.002 | 2.679  |

|    |            |     |               |        |      |        |       |
|----|------------|-----|---------------|--------|------|--------|-------|
| 07 | mechanical | yes | switch budget | 17.996 | 18.0 | 15.000 | 2.361 |
| 08 | mechanical | no  | dry-bulb      | 20.194 | 18.0 | 14.440 | 2.792 |
| 09 | mechanical | no  | dry-bulb      | 24.686 | 18.0 | 10.853 | 3.863 |
| 10 | mechanical | no  | dry-bulb      | 28.431 | 18.0 | 6.823  | 4.999 |
| 11 | mechanical | no  | dry-bulb      | 28.950 | 18.0 | 4.593  | 5.850 |
| 12 | mechanical | no  | dry-bulb      | 27.770 | 18.0 | 3.459  | 5.988 |
| 13 | mechanical | no  | dry-bulb      | 27.432 | 18.0 | 2.782  | 5.782 |
| 14 | mechanical | no  | dry-bulb      | 25.184 | 18.0 | 1.672  | 4.861 |
| 15 | mechanical | no  | dry-bulb      | 19.592 | 18.0 | 1.672  | 4.428 |
| 16 | FREE       | yes | -             | 13.354 | 18.0 | 0.689  | 3.351 |
| 17 | FREE       | yes | -             | 9.160  | 18.0 | 0.153  | 3.375 |
| 18 | FREE       | yes | -             | 6.182  | 18.0 | -0.707 | 2.793 |
| 19 | FREE       | yes | -             | 4.642  | 18.0 | -1.237 | 3.208 |
| 20 | FREE       | yes | -             | 2.156  | 18.0 | -1.787 | 3.490 |
| 21 | FREE       | yes | -             | 1.768  | 18.0 | -2.347 | 3.839 |
| 22 | FREE       | yes | -             | 0.709  | 18.0 | -3.465 | 4.289 |
| 23 | FREE       | yes | -             | -0.097 | 18.0 | -3.796 | 4.328 |

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin  
actual = what the intake temperature turned out to be, from the station record  
margin = group-conditional forecast error for that hour of day, plus plume spread

## THE REASONING, HOUR BY HOUR

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### 00:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.809 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

### 01:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.805 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

### 02:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.319 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

### 03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.403 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.319 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

### 05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.285 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is

also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**06:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.952 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**07:00 MECHANICAL [switch budget]**

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.996 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

**08:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 20.194 C against a 18.0 C limit, so it fails by 2.194 C. It would take a limit 2.194 C higher, or a bound 2.194 C tighter, to change this hour.

**09:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 24.687 C against a 18.0 C limit, so it fails by 6.687 C. It would take a limit 6.687 C higher, or a bound 6.687 C tighter, to change this hour.

**10:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 28.431 C against a 18.0 C limit, so it fails by 10.431 C. It would take a limit 10.431 C higher, or a bound 10.431 C tighter, to change this hour.

**11:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 28.949 C against a 18.0 C limit, so it fails by 10.949 C. It would take a limit 10.949 C higher, or a bound 10.949 C tighter, to change this hour.

**12:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 27.770 C against a 18.0 C limit, so it fails by 9.770 C. It would take a limit 9.770 C higher, or a bound 9.770 C tighter, to change this hour.

**13:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 27.432 C against a 18.0 C limit, so it fails by 9.432 C. It would take a limit 9.432 C higher, or a bound 9.432 C tighter, to change this hour.

**14:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 25.184 C against a 18.0 C limit, so it fails by 7.184 C. It would take a limit 7.184 C higher, or a bound 7.184 C tighter, to change this hour.

**15:00 MECHANICAL [dry-bulb]**

Mechanical. The upper bound on intake air is 19.592 C against a 18.0 C limit, so it fails by 1.592 C. It would take a limit 1.592 C higher, or a bound 1.592 C tighter, to change this hour.

**16:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 13.354 C, which is 4.646 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.351 C, and the margin is measured, not chosen: 3.102 C of group-conditional forecast error for this hour of day, plus 0.2483 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

**17:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 9.160 C, which is 8.840 C

under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.375 C, and the margin is measured, not chosen: 3.096 C of group-conditional forecast error for this hour of day, plus 0.2789 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

#### **18:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 6.182 C, which is 11.818 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.793 C, and the margin is measured, not chosen: 2.494 C of group-conditional forecast error for this hour of day, plus 0.2990 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

#### **19:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 4.642 C, which is 13.358 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.208 C, and the margin is measured, not chosen: 2.880 C of group-conditional forecast error for this hour of day, plus 0.3279 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

#### **20:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 2.156 C, which is 15.844 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.490 C, and the margin is measured, not chosen: 3.181 C of group-conditional forecast error for this hour of day, plus 0.3094 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

#### **21:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 1.768 C, which is 16.232 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.839 C, and the margin is measured, not chosen: 3.527 C of group-conditional forecast error for this hour of day, plus 0.3127 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

#### **22:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is 0.709 C, which is 17.291 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.289 C, and the margin is measured, not chosen: 3.973 C of group-conditional forecast error for this hour of day, plus 0.3163 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

#### **23:00 FREE-COOLING**

Free cooling. The 90 %-nominal upper bound on intake air is -0.097 C, which is 18.097 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.328 C, and the margin is measured, not chosen: 4.027 C of group-conditional forecast error for this hour of day, plus 0.3009 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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